

Effect of the Traditional Korean Immunomodulating Formulation, Gamguntang (GGT), on Experimental Thyroiditis Model

Bong-Seok Kang,¹ Ji-Yeong Han,² Sung-Koo Kang,¹ Dong-IL Kim,² June-Ki Kim,² and Cheorl-Ho Kim¹

¹Department of Biological Sciences, Sungkyunkwan University, Suwon City, Kyunggi-Do, Korea

²Department of Oriental Medicine, Dongguk University and MRC for Cardiovascular Natural Products, Kyungju, Kyungbuk, Korea

The crude herbal formulation, Gamguntang (GGT), is an immunomodulator showing marked down-regulation of several experimental autoimmune diseases. In this study, its effect on different experimental models of thyroid disease was investigated. Although very effective at preventing thyroid infiltrates in mice immunized with mouse deglycosylated thyroglobulin and complete Freund's adjuvant and in spontaneous models of thyroiditis, it completely failed to modify experimental autoimmune thyroiditis (EAT) induced in mice immunized with mouse thyroglobulin and lipopolysaccharide. There was no significant shift in the observed isotypes of anti-mouse thyroglobulin antibodies and only anti-mouse thyroglobulin antibodies in the spontaneous model were completely down-modulated by the GGT. One surprising fact to emerge was that GGT-treated donor mice, although protected from thyroid lesions themselves, were still able to transfer EAT showing that they must have been effectively primed while being treated with GGT. It is possible that the drug down modulated EAT by interfering with the trafficking of primed effector cells.

Keywords Autoimmunity, Tel, Cytotoxic T lymphocyte, *In vivo* animal model, oriental herbal extract, Gamguntang, Glycyrrhizae Radix (*Glycyrrhiza uralensis* Fischer; *Glycyrrhiza Fissch*), Black bean (*Glycine max* L. Merr), Angelicae radix (*Angelica gigas* Nakai), Cnidii Rhizoma (*Cnidium officinale* Makino).

Address correspondence to Cheorl-Ho Kim, Department of Biological Sciences, Sungkyunkwan University, Chunchun-Dong 300, Jangan-Gu, Suwon City, Kyunggi-Do 440-746, Korea. Fax: +82-31-290-7015. E-mail: chkimbio@skku.edu

INTRODUCTION

Autoimmune thyroid disease affects up to 1% of the population and is characterized by the presence of infiltrating lymphocytes and circulating antibodies to thyroid antigens. Thyrocyte destruction and thyroid atrophy are obvious features of autoimmune thyroid disease, occurring in both human and experimental thyroiditis, but the mechanisms responsible are not fully understood. A number of observations suggest a crucial participation of lymphokines in pathology of autoimmune thyroiditis. It was shown that mitogen-stimulated T cell clones derived from thyroid glands from Hashimoto's thyroiditis (HT) patients secrete predominantly interferon (IFN)- γ ⁽¹⁾ and tumor necrosis factor (TNF)- α .^(2,3) Another recently described feature of autoimmune thyroid disease is an increased frequency of major histocompatibility complex (MHC) class II antigens on some of the thyroid cells in the affected gland.⁽⁴⁾ This aberrant expression can render the thyroid follicular cell capable of presenting autoantigens to the immune system^(4,5) and initiate or perpetuate the autoimmune response.

There is a spectrum of human autoimmune thyroiditis in which antibodies and T cells contribute differentially to the pathology.⁽⁶⁾ A variety of animal models for experimental autoimmune thyroiditis (EAT) exist which may arise as a result of different mechanisms. One model extensively employed involves the immunization of susceptible mice (H-2^k) with autologous thyroglobulin in complete Freund's adjuvant (CFA) or lipopolysaccharide (LPS).^(7,8) In this model, mononuclear cells consisting of both CD4+ and CD8+ T cells, together with macrophages and a few B cells,⁽⁹⁾ infiltrate the gland. There has been much speculation on the relative contribution of T helper type I (Th1) and Th2 subsets in this model, and there is some evidence that the EAT induced in normal CBA/J (I-A^k) is characteristic of a Th1 response.⁽¹⁰⁾ However, It was also reported that the Th2 subset may be involved, as in their model of granulomatous EAT, inhibiting the Th1 subset with anti-interferon- γ (IFN- γ) antibody did not prevent thyroid infiltration.⁽¹¹⁾ Our recent study have also evaluated the contribution of these CD8+ CTL cells to the pathogenesis of EAT by immunization of naive CBA/J mice with N-glycanase F and trifluoromethanesulfonic acid (TMSF)-deglycosylated porcine thyroglobulin (dgpTg) without the aid of adjuvants. dgpTg-immunized mice developed a different EAT from conventional EAT induced by thyroglobulin (Tg) and adjuvants, i.e., cytotoxic responses to dgpTg were detected in T cells from dgpTg-immunized mice.⁽¹²⁾

Another murine model of thyroiditis has recently been developed which may be more akin to the spontaneously arising human disease. Wicker and colleagues have derived a non-obese diabetic (NOD) congenic mouse (NOD H-2^{h4}) bearing I-A^k, which spontaneously develops EAT but no insulin-dependent diabetes mellitus (IDDM). This EAT can be accelerated and enhanced by the addition of iodine to the drinking water.^(13,14) The relative

importance of Th1 and Th2 subsets in this model has not been established. Rose and colleagues found the anti-mouse thyroglobulin (MTg) antibodies in this mouse model were predominantly immunoglobulin G2a (IgG2a) and IgG2b with no IgG1,⁽¹⁴⁾ whereas Damotte et al. using an independently derived cross of NOD and B10BR mice, found IgG2a poorly represented in these animals.⁽¹⁰⁾ NOD mice (I-A^{g7}), as well as developing insulinitis leading to IDDM, are also prone to develop inflammatory infiltrates in the salivary gland in both sexes.⁽¹⁵⁾ The NOD mouse can therefore be considered a model for experimental thyroiditis.

Gamguntang (GGT), which is consisted of 4 medicinal herbs such as Glycyrrhizae Radix (*Glycyrrhiza uralensis* Fischer; *Glycyrrhiza Fissch*), Black bean (*Glycine max* L. Merr), Angelicae radix (*Angelica gigas* Nakai) and Cnidii Rhizoma (*Cnidium officinale* Makino), has been shown to stimulate several immune functions and has been found to be effective in the prevention of cancer metastases.⁽¹⁶⁾ It also modifies a number of animal models of autoimmune diseases, including experimental autoimmune encephalomyelitis (EAE) and collagen-induced arthritis (CIA).⁽¹⁷⁾ They also showed a strong cytotoxicity to human A549 lung carcinoma cells, cervix epithelioid carcinoma Hela cells, and Hepa 1c1c7 hepatoma cells with approximately 80% growth inhibition.⁽¹⁸⁾ GGT increased the proliferation of T and B cells. In addition, GGT remarkably increased cytolytic activity of purified T cell against K562 cells, indicating its immunomodulating effect.⁽¹⁹⁾ Recently, it was shown that GGT protects cytokine-induced cytotoxicity and MHC class II antigen expression *in vitro* in the continuously growing and differentiated thyroid cell line, FRTI.⁽²⁰⁾ GGT shows no evidence of general immunosuppression, and the convenience of oral administration has made it a drug of considerable interest in the treatment of autoimmune disorders.

In this study, we have studied the effect of GGT on the experimental thyroiditis induced by the immunization of CBA/J mice (H-^{2k}) with deglycosylated pig thyroglobulin using two separate adjuvants (CFA and LPS).

MATERIALS AND METHODS

Mice

CBA/J female and male mice were purchased from KRIBB. All mice were 6–8 weeks of age at the time of use.

Materials

Con-A Sepharose (particle size: 45–165 μ m, 10–16 mg Con-A per gel sediment) was obtained from Pharmacia (Uppsala, Sweden). Calibration proteins for molecular weight determination were obtained from Pharmacia. pTg was from Sigma Chemical Co. (St. Louis, MO, USA) and other chemicals were of analytical reagent grade and obtained from Merck (Darmstadt, F.R.G.).

Preparation of GGT

GGT was dissolved in tap water at a concentration of 200 mg/L to obtain an approximate dose of 20 mg/kg/day. The pH was then adjusted to 7.0. Controls were given normal tap water. GGT was added *in vitro* at 100 µg/mL.⁽¹⁶⁾

Preparation of Mouse Thyroglobulin

This was obtained by extraction from homogenized thyroid tissue in PBS and subsequent differential ammonium sulphate precipitation (35–45%) followed by extensive dialysis against phosphate-buffered saline (PBS). The thyroglobulin was stored in frozen aliquots until required.

Preparation of dgpTg

Porcine Tg (pTg) from Sigma (St. Louis, MO, USA) was diluted in distilled water and denatured by boiling in a microwave oven until a visible precipitation was observed.⁽²¹⁾ Deglycosylated porcine Tg (dgpTg) was characterized by chromatography and electrophoresis (Fig. 1) as described.⁽¹²⁾ Native pTg or dgpTg (50 µg) was electrophoresed on a 5–25% (wt/vol) acrylamide gel and 0.1% (wt/vol) SDS with a Tris-glycine buffer system by the method of Laemmli.⁽²²⁾ Twenty mg dgpTg or pTg in 500 µL distilled water

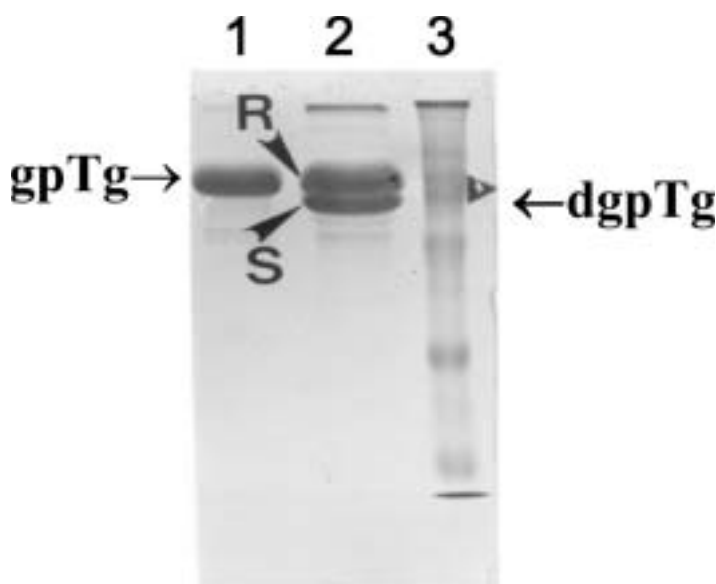


Figure 1: SDS-PAGE analysis of the purified pTg, after Endo F digestion (lane 2) in comparison with the untreated pTg (lane 1). Lane 3, standard molecular weight. R, resistant protein (about 330 kDa) against Endo F digestion; S, sensitive proteins (about 250 kDa) against Endo F digestion allowing effective deglycosylation.

were loaded onto Sepharose 4B gel, whereas pTg eluted as two peaks of 660 and 330 kDa, respectively, dgpTg eluted as two peaks of about 600 and 300 kDa. SDS-PAGE of pTg also showed a band at 330 kDa, whereas most of dgpTg was located in a band at 250 kDa (Fig. 1). These data suggested that deglycosylation induced aggregation of pTg.

After electrophoresis, the gel was stained with coomassie blue. After endo- β -N-acetylglucosaminidase F (Endo F) and trifluoromethane sulfonic acid (TFMS) experiments, SDS-PAGE were loaded with protein amounts allowing only the visualization, after Coomassie staining, of the major deglycosylated Tg. This avoided difficulties of interpretation due to the presence of minor Tg forms, which are not fully deglycosylated. Electrophoresis of dgpTg showed two bands at 330 and 660 kDa, whereas most of native pTg was located in a band at 660 kDa.⁽¹²⁾ Digestion of the pTg by Endo F yielded one reaction product to the glycosidase action. The product differed from the original pTg in molecular weight by about 8000 Da. This, in conjunction with the experiment of Cona and WGA binding indicated the presence of high mannose N-linked oligosaccharides in the pTg.

Before Con A or WGA/alkaline phosphatase staining, proteins were transferred from SDS-gels to nitrocellulose paper as previously described.⁽²³⁾ N-linked oligosaccharides were characterized by digoxigenin specific antibody conjugated with alkaline phosphatase (DIG Glycan detection Kit, Boehringer Mannheim, GmbH, Germany) and affino blotting with Con A/Alkaline phosphatase (Lectin Con A-biotin-labeled, BM) or WGA/alkaline phosphatase (Lectin WGA-biotin-labeled, BM) as described by Hawkes⁽²⁴⁾ but with skim with milk as a blocking agent. Enzymatic removal of the high mannose glycans with Endo-F was performed according to Trimble and Maley.⁽²⁵⁾ Tg (2 mg protein) was deglycosylated by 5.0 units Endo F (Boehringer mannheim) in 400 μ L of 25 mM acetate buffer (pH 5.0) at 37°C for 12 h. Deglycosylation with TFMS was carried out according to Edge et al.⁽²⁶⁾ The molecular weight of deglycosylated Tg was estimated by FPLC Sepharose-4B gel filtration and SDS-PAGE.

Immunization with Tg and dgpTg

Injections were performed i.v. on day 0 and 14, with 100 μ g dgpTg in saline without adjuvant. Simultaneously, pTg was emulsified in complete freund adjuvant (CFA) for immunization on day 0 and in incomplete freund adjuvant (IFA) for challenge on day 14. CFA suspension, which contained 1 mg/mL *Mycobacterium tuberculosis* (Difco Lab, Detroit, MI, USA), was injected s.c. with 100 μ g pTg. Control mice were injected with adjuvant suspensions alone (s.c) or saline (i.v.).

Adjuvant-induced EAT

Mice were immunized twice, one week apart with 50 μ g deglycosylated pTg emulsified in CFA (H37Ra Difco Laboratories, Detroit, MI, USA). The first immunization was subcutaneous (s.c.) at the base of the tail, and the second

was divided between the two hind footpads. The mice were bled and thyroids were taken for histology on day 14 or day 21. Alternatively, mice were given 50 µg dgpTg intravenously (i.v.) followed 3 hr later by 20 µg LPS (*Salmonella enteritidis* Sigma Chemical Co., St. Louis, MO, USA) given i.v. This was repeated on day 7 and the mice were bled and thyroids were taken on day 28.

Sodium Iodide-induced EAT

Mice were given tap water containing 0.05% iodine obtained from sodium iodide (Sigma). This represents an increase over normal intake of 550 fold, and mice were maintained on this for 6 weeks. They did not appear to find the water distasteful since similar quantities to normal tap water were consumed daily.

Transfer of Thyroiditis

Donor mice were immunized with dgpTg/CFA on day 0 and 7 as described above and the popliteal lymph nodes were taken on day 15. Cells from these nodes were cultured for 3 days in Petri dishes (NUNC; Nalge Nunc International, Roskilde, Denmark) at 5×10^6 /mL in RPMI-1640 (Gibco, Bethesda, MD, USA) with 5% fetal calf serum (FCS) and other standard supplements⁽²⁷⁾ and dgpTg 50 µg/mL. This allowed the activation and expansion of the effector T cells. After washing, 2×10^7 viable cells were injected i.v. into normal syngeneic recipients.

Evaluation of Thyroid Infiltration

Thyroids were fixed in formal saline and sections were stained with hematoxylin-eosin. There were 12 scores possible for each mouse (two lobes at six levels): 0, no infiltration; 1, any definite infiltration up to 10% of the gland; 2, between 10 and 40%; 3, between 40 and 80%; 4, gland totally infiltrated but follicles still discernible; 5, no follicles distinguishable. The sections are scored blind, and the total for each mouse is divided by the number of observations for that mouse. The average thyroiditis for each group is the mean $\pm$ SE of all the mice in the group.

Enzyme-linked Immunosorbent Assays

Anti-thyroglobulin autoantibodies were assayed using standard ELISA methods.⁽²⁸⁾ The plates were coated overnight at 4°C with mouse thyroglobulin at 10 µg/mL. A goat anti-mouse polyvalent immunoglobulin (IgM, IgG and IgA) conjugated to alkaline phosphatase (Sigma Immunochemicals) was used as developing antibody. The values shown are the mean optical density readings at 410 nm $\pm$ SE. Iso-types were evaluated by using anti-mouse IgG1, -2a or -2b at 1/1500 (Sigma).

Statistics

Students *t*-test was used where appropriate.

RESULTS

Characterization of hdpTG

dgpTg was characterized by chromatography and electrophoresis. Twenty mg dgpTg or pTg in 500 μ l distilled water were loaded onto Sepharose 4B gel, whereas pTg eluted as two peaks of 660 and 330 kDa, respectively, dgpTg eluted as two peaks of about 600 and 300 kDa. SDS-PAGE of pTg also showed a band at 330 kDa, whereas most of dgpTg was located in a band at 250 kDa (Fig. 1). These data suggested that deglycosylation induced aggregation of pTg. Endo F is known to specifically remove the high-mannose N-linked oligosaccharides. After enzymatic digestion of the pTg, products were analyzed by SDS-PAGE. The molecular weight was noted for the pTg (Fig. 1). Digestion of the pTg by Endo F yielded one reaction product to the glycosidase action. The product differed from the original pTg in molecular weight by about 8000 Da. This, in conjunction with the experiment of ConA and WGA binding, indicated the presence of high mannose N-linked oligosaccharides in the pTg.

GGT Effectively Blocked the Development of EAT Induced by dgpTg/CFA

The drug was administered in tap water to CBA/J mice at ~20 mg/kg a day as described in the Materials and Methods and was started at day -3, day 0, or day 3 of the experiment. On day 0, the mice were immunized with 50 μ g dgpTg/CFA in the base of the tail, and this dose was repeated on day 7 but administered in the hind footpads. Control mice were also immunized but were maintained on normal tap water. All mice were bled and the thyroids were taken at day 14. Fig. 2A shows that the GGT significantly abrogated EAT when given just before the first immunizing dose of dgpTg, and the anti-dgpTg autoantibodies, which are predominantly IgG1 and IgG2b but with IgG2a well represented, were only moderately reduced (Fig. 2B). However, GGT slightly abrogated EAT without any significance when given just after the first immunizing dose of dgpTg and the anti-dgpTg autoantibodies (Fig. 2A). Subsequent experiments confirmed these results and isotypic analysis revealed a very modest reduction or none in all IgG subclasses (Fig. 3).

GGT was Only Effective When Administered Soon After Immunization and Had to be Given Continuously

GGT started 1–3 days after the first priming dose of dgpTg prevented EAT, but if delayed until the boosting dose on day 7, it was ineffective. In a separate experiment, CBA/J mice were treated with GGT for 4 weeks and then primed with dgpTg/CFA either immediately after stopping the GGT or 1 week later.

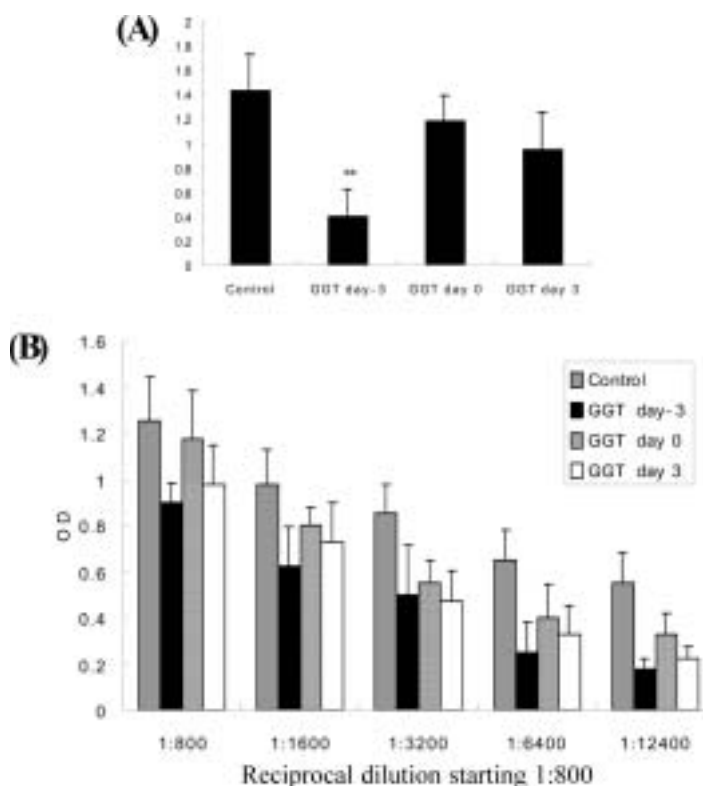


Figure 2: CBA/J mice were immunized with 50 μ g dgpTg/CFA s.c. at the base of the tail on day 0 and boosted with the same dose s.c. in the hind of footpads on day 7. GGT was administered in the drinking water at an approximate dose of 200 mg/kg/day beginning on the days indicated. Mice were bled and thyroids taken on day 14. (A) The thyroiditis scored according to the criteria given in the methods section (** $p \leq 0.01$ for GGT day -3 group compared with controls). (B) The anti-dgpTg antibodies evaluated in an ELISA using goat anti-mouse polyvalent immunoglobulin conjugated to alkaline-phosphatase as second antibody ($P \leq 0.01$ at 1/1600 and 1/6400 for all groups compared with controls.) All values are the means $\pm$ SE of seven mice in each group.

GGT failed to prevent EAT in both these groups. When a short pulse of GGT was given for the first 7 days following priming and then administration was discontinued, no difference in the subsequent thyroid infiltrates was seen between controls and those mice given a 7-day course of GGT. These studies demonstrated that GGT must be administered continuously to be effective and indicate that GGT can only prevent this model of EAT if it is administered within 3 days of priming and then continuously for the entire induction phase.

GGT was not Effective in the Transfer System

EAT may also be induced in CBA/J mice by the i.v. transfer of primed cells taken from donor draining lymph nodes 7 days after the boosting immunization

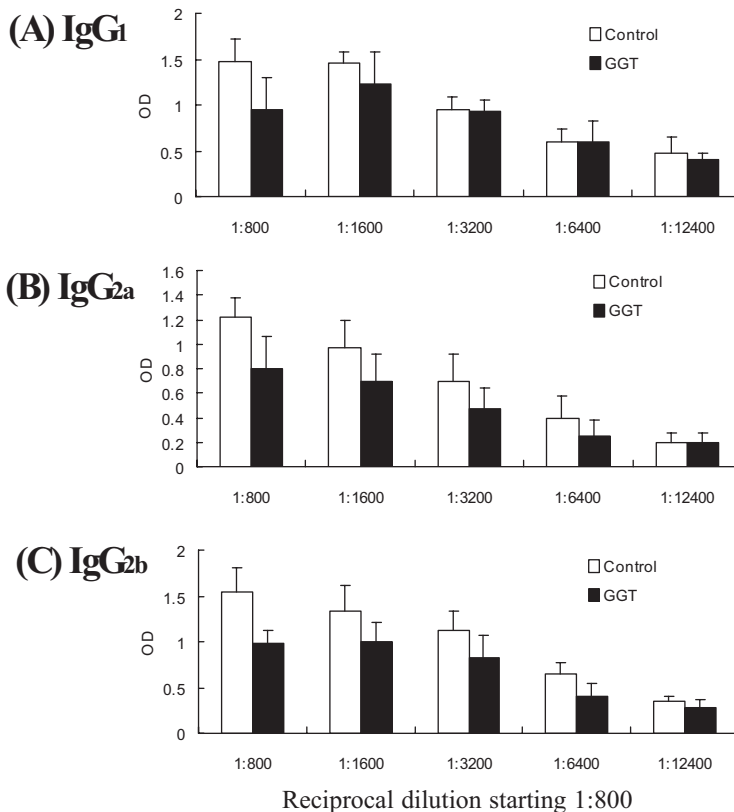


Figure 3: CBA/J mice were immunized twice with 50 μ g dgpTg/CFA as before $\pm$ GGT in the drinking water. The mice were bled on day 14 and the isotype profile was examined using alkaline-phosphatase-conjugated anti-IgG1, IgG2a and IgG2b antibodies for detection.

in the footpad described above. To be effective, these cells must be restimulated with dgpTg *in vitro* for 72 hr before transfer and, consequently, the system affords several different opportunities for intervention. It is possible to address the following questions: 1) can transfer be prevented by putting GGT into the culture; 2) can it be prevented by treating the recipients of normal primed cells with GGT; and finally; 3) can GGT-treated donor mice still transfer the disease? With reference to the first two questions, we found that neither putting GGT in culture with the stimulating antigen nor treating the recipient mice with GGT from 3 days before transfer had any effect on the EAT mediated by primed cells from normal donors (Fig. 4). It is possible therefore, that a longer period of pretreatment may have been more effective in our EAT transfer model.

Draining lymph node cells from donor mice treated with GGT right up until death were surprisingly still able to transfer thyroiditis in three-quarters of recipients (Fig. 5). Macroscopic examination of donor thyroids at death revealed

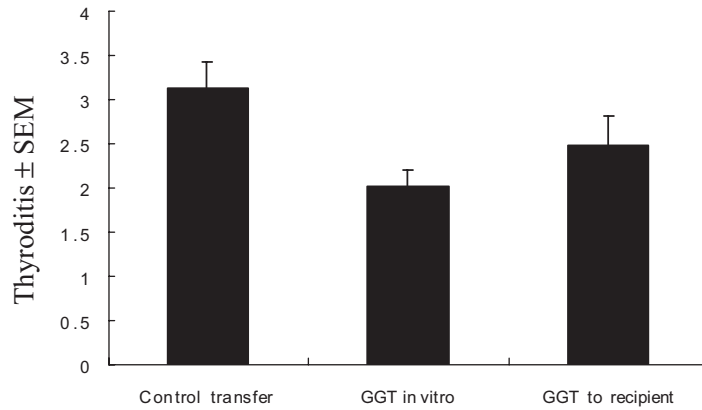


Figure 4: Donor CBA/J mice were immunized with 50 µg dgpTg/CFA on days 0 and 7 as before and on day 14, the popliteal lymph nodes were removed and cultured as a single cells suspension for 3 days with dgpTg 50 µg/mL. After washing, 2×10^7 viable cells were transferred i.v. into normal syngeneic recipients. GGT was either added to the culture medium at 100 µg/mL or added to the drinking water of the recipients starting 3 days before transfer. Thyroids were examined 14 days after transfer and scored as before. Values shown are mean $\pm$ SE of five mice in each group.

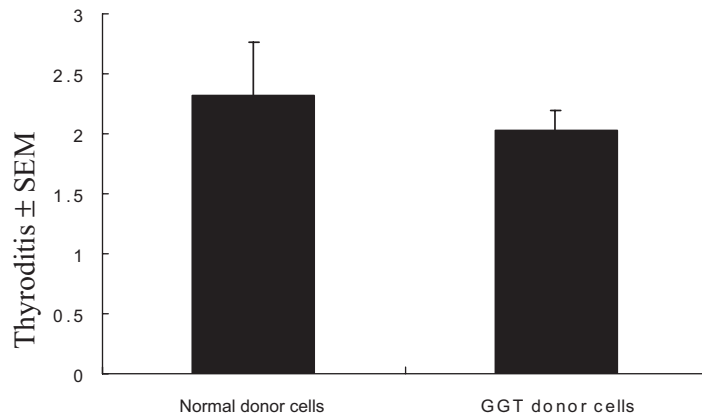


Figure 5: Donor CBA/J mice were immunized with 50 µg dgpTg/CFA on days 0 and 7 $\pm$ GGT in the drinking water starting 3 days before the first immunization. Popliteal lymph nodes were removed on day 14 and cultured with dgpTg 50 µg/mL but without GGT, for 3 days. After washing, 2×10^7 viable cells were transferred i.v. to normal syngeneic recipients. The thyroids were examined after 14 days and the superimposed values shown are individual.

that thyroids from GGT-treated mice appeared to be completely normal whereas the control donors displayed obvious EAT. To see if GGT could be affecting trafficking of primed cells to the thyroid, we stained sections from the draining lymph nodes for the following adhesion molecules, platelet endothelial cell adhesion molecule (PECAM)-1, vascular cell adhesion molecule (VCAM)-1,

mucosal addressin cell adhesion molecule (MAdCAM)-1 and intracellular adhesion molecule-1, lymphocyte function-associated antigen-1 and very late activation antigen-4. Preliminary experiments showed no demonstrable differences between normally primed mice and mice treated with GGT.

GGT is Equally Effective in Other Animal Models of EAT

dgpTg administered i.v. (50 μ g) with LPS (20 μ g) as accompanying adjuvant is another effective method for inducing infiltration of the thyroid gland and circulating autoantibodies in CBA/J mice.⁽¹²⁾ The infiltrates are usually modest and may differ in cellular composition from those induced using CFA. When mice maintained on GGT since day -3 were immunized using the LPS protocol, we found the drug had little or no significant effect on the development of thyroid infiltration (Fig. 6) and anti-dgpTg autoantibodies or on the isotypes represented (data not shown). Like the human disease, it is age related and progressive and iodine intake appears to have some bearing on its incidence and severity.⁽²⁹⁾

DISCUSSION

Most existing treatments for autoimmunity are based on nonspecific immunosuppression or antiinflammatory agents and although often very effective at down-regulating the relevant autoimmune responses may also lead to increased susceptibility to other complications associated with depressed immune function. EAT was induced by i.v. injection of deglycosylated senogenic Tg without the aid of adjuvants. Denaturation of exogenous proteins prior to immunization was achieved by deglycosylation of pTg, which was demonstrated by electrophoresis, chromatography and glycan analysis using DIG-oxigenin

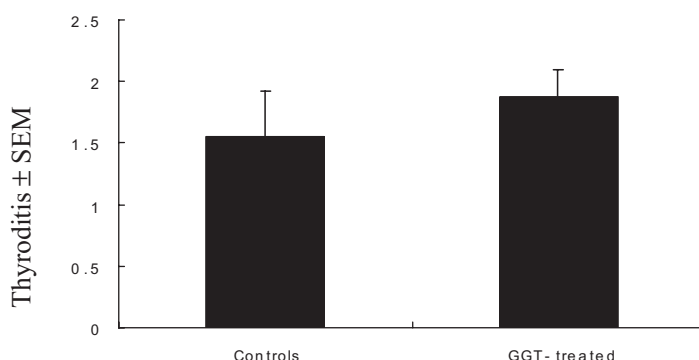


Figure 6: CBA/J mice were maintained $\pm$ GGT and immunized on days 0 and 7 with dgpTg followed 3 h later by 20 μ g LPS, both given i.v. Thyroids were taken on day 21 and the scores shown represent the mean $\pm$ SE of five mice in each group.

labelling kit to induce its aggregation. The deglycosylated 660-kDa Tg was comparable to the 660-kDa globules of Tg isolated from bovine thyroid glands,⁽³⁰⁾ which represents a form of storage of highly concentrated Tg (100–400 mg/mL). Release of such Tg globules from normal thyroid glands of rats was observed *in vivo*, following specific signals for exocytosis.⁽³¹⁾

GGT is consisted of 4 medicinal herbs of *Glyrrhiza uralensis* Fischer, *Glycine max* L. Merr, *Angelica gigas* Nakai and *Cnidium officinale* Makino, and it has long been used for the prevention of cancer metastases⁽¹⁶⁾ and for the autoimmune diseases including experimental autoimmune encephalomyelitis (EAE) and collagen-induced arthritis (CIA).⁽¹⁷⁾ GGT has also potent inhibitory effects on human A549 lung carcinoma cells, cervix epithelioid carcinoma Hela cells and Hepa1c1c7 hepatoma cells.⁽¹⁸⁾ In addition, GGT increased the immune-stimulating activities on T and B cells with cytolytic activity against K562 cells, indicating its immunomodulating effect.⁽¹⁹⁾ Therefore, GGT has shown exceptional promise as an immunomodulator having no generally immunosuppressive effects but inducing marked amelioration of several autoimmune diseases. Furthermore, it protects cytokine-induced cytotoxicity and MHC class II antigen expression *in vitro* in the thyroid cells.⁽²⁰⁾ GGT shows no evidence of general immunosuppression, and the convenience of oral administration has made it a drug of considerable interest in the treatment of autoimmune disorders. With the above background, therefore, we have studied the effect of GGT on the experimental thyroiditis induced by the immunization of CBA/J mice (H-2^k) with deglycosylated pig thyroglobulin using two separate adjuvants (CFA and LPS). Our data show that in common with other experimental models of autoimmunity, EAT is readily down-modulated by GGT. However, its effectiveness in EAT was restricted since we found that GGT must be administered at, or very soon after immunization, and must be given continuously. Curiously, thyroiditis, induced by dgpTg with LPS as adjuvant, appeared to be unaffected by GGT, suggesting the underpinning mechanisms of induction are different with CFA and LPS. It is difficult to explain how this could be as there is evidence that LPS primes T cells⁽³²⁾ as indeed it must since EAT is a T-cell-mediated disease.

There are several levels at which it may be envisaged that GGT exerts its modulating effect on EAT. We wish to consider three of these possibilities. A antigen presentation, cell trafficking and alteration in subset bias. It has been proposed that GGT may cause an upregulation of natural killer (NK) cells, which subsequently kill dendritic cells following their interaction with antigen.⁽³³⁾ Our data do not lend support to the notion that GGT mediates its effect via antigen presentation as GGT-treated donors were clearly primed since they were able to transfer EAT. If, however, GGT prevents trafficking from the lymph node, this could be one explanation for the observed data since although primed T cells might be trapped *in situ* in donor mice, transfer of these cells intravenously would release them in the recipient. Although it is

probable that more extensive pretreatment of recipients is required to prevent transfer of EAT, it is still unlikely that there would be no observed effect after even only 3 days pretreatment if antigen presentation was being targeted by GGT in this system. If on the other hand, GGT only blocks egress of T cells from the lymph node draining the site of immunization, no effect would be expected in GGT-treated recipients. Several workers have shown that interfering with trafficking molecules can protect animals from disease in experimental systems.^(34,35) If GGT does indeed block egress from lymph nodes, this would also explain why it was ineffective in the LPS model where intravenous immunization results in antigen-specific lymphocytes being present in the spleen. We were, however, unable to provide any evidence to support this theory since there was no appreciable difference in the expression of those adhesion molecules examined in the draining lymph nodes from experimental and control animals. It is possible that a more extensive study of such molecules could reveal some differences. Alternatively, it could be argued that the *in vitro* step before transfer releases primed cells from the effects of GGT allowing them to express the homing molecules required for trafficking to the thyroid in recipient mice.

As it has been shown that GGT upregulates a number of immune functions like interleukin-2 and NK cells (not shown), it may be that it enhances T-cell activation in autoimmune situations to the point at which damage limiting mechanisms are activated. It has also been shown that GGT may differentially depress cytokine production from Th1 and Th2 subsets (not shown) and this could provide another means by which the drug might down-regulate EAT. However, our data show no shift in antibody isotype profile following GGT treatment. IgG1, IgG2a, and IgG2b were all well represented in both adjuvant models and any reduction after GGT was similar for all three isotypes. Finally, our data indicate that intervention therapy, although promising, should be approached cautiously since a knowledge of the underlying mechanism may be crucial for predicting the success or otherwise of GGT treatment under different conditions.

ACKNOWLEDGMENT

This work was supported by Vascular and Natural Products MRC, KOSEF, Korea.

ABBREVIATIONS

GGT	Gamguntang
HT	Hashimoto's thyroiditis
IFN	interferon

TNF	tumor necrosis factor
MHC	major histocompatibility complex
CTL	cytotoxic T lymphocyte
EAT	experimental autoimmune thyroiditis
Tg	thyroglobulin
Endo-r	endo- β -N-acetylglucosaminidase F
TMSF	trifluoromethanesulfonic acid
pTg	porcine thyroglobulin
dgp	Tg,deglycosylated porcine Tg
Tc1	type 1 cytotoxic T cell
p.i.	post immunization

REFERENCES

1. Del Prete, G.F.; Tiri, A.; Mariotti, S.; Pinchera, A.; Ricci, M.; Romagnani, S. Enhanced production of gamma-interferon by thyroid-derived. T cell clones from patients with Hashimoto's thyroiditis. *Clin. Exp. Immunol.* **1987**, *69*, 323–337.
2. Del Prete, G.F.; Tiri, A.; De Carli, M.; Mariotti, S.; Pinchera, A.; Chretien, I.; Romagnani, S.; Ricci, M. High potential to tumor necrosis factor alpha (TNF-alpha) production of thyroid infiltrating T lymphocytes in Hashimoto's thyroiditis: a peculiar feature of destructive thyroid autoimmunity. *Autoimmunity* **1989**, *4*, 267–276.
3. Turner, M.; Londei, M.; Feldmann, M. Human T cells from autoimmune and normal individuals can produce tumor necrosis factor. *Eur. J. Immunol.* **17**, 1807–1814.
4. Hanafusa, T.; Pujol-Borrell, R.; Chiovato, L.; Russell, R.C.G.; Doniach, D.; Bottazzo, G.F. Aberrant expression of HLA-DR antigen on thyrocytes in Graves' disease: relevance for autoimmunity. *Lancet*, **1983**, *2*, 1111–1115.
5. Bottazzo, G.F.; Pujol-Borrell, R.; Hanafusa, T.; Feldmann, M. Role of aberrant HLA-DR expression and antigen presentation in induction of endocrine autoimmunity. *Lancet*. **1983**, *2*, 1115–1119.
6. Dawe, K.; Hutchings, P.R.; Champion, B.; Cooke, A.; Roitt, I.M. Autoantigens in thyroid disease. *Springer Semin Immunopathol.* **1993**, *14*, 285–307.
7. Esquivel, P.S.; Rose, N.R.; Kong, Y.C.M. Induction of autoimmunity in good and poor responder mice with mouse thyroglobulin and lipopolysaccharide. *J. Exp. Med.* **1977**, *145*, 1250–1255.
8. Hutchings, P.R.; Cooke, A.; Dawe, K.; Champion, B.R.; Geysen, M.; Valerio, R.; Roitt, I.M. A thyroxine-containing peptide can induce murine experimental autoimmune thyroiditis. *J. Exp. Med.* **1992**, *175*, 869–872.
9. Flynn, J.C.; Conaway, D.H.; Cobbold, S.; Waldmann, H.; Kong, Y.C.M. Depletion of L3T4+ and Lyt-2+ cells by rat monoclonal antibodies alters the development of adoptively transferred experimental autoimmune thyroiditis. *Cell. Immunol.* **1989**, *122*, 377–390.
10. Damotte, D.; Colomb, E.; Cailleau, C.; Brousse, N.; Charreire, J.; Carnaud, C. Analysis of susceptibility of NOD mice to spontaneous and experimentally induced thyroiditis. *Eur. J. Immunol.* **1997**, *27*, 2854–2862.

11. Stull, S.J.; Sharp, G.C.; Kyriakos, M.; Bickel, J.T.; Braley-Mullen, H. Induction of granulomatous experimental autoimmune thyroiditis in mice with in vitro activated effector T cells and anti-IFN-antibody. *J. Immunol.* **1992**, *149*, 2219–2226.
12. Kim, Y.K.; Kang, B.S.; Chung, T.W.; Moon, J.Y.; Lim, J.K.; Shon, Y.H.; Nam, K.S.; Kim, D.S.; Jeon, B.H.; Kim C.H., Induction of a Tc1-mediated experimental autoimmune thyroiditis by deglycosylated porcine thyroglobulin. *Immunopharm. Immunotoxicol.* **2004**,
13. Podolin, P.L.; Pressey, A.; Delaroto, N.H.; Fischer, P.A.; Peterson, L.B.; Wicker, L.S. I-E+ Nonobese diabetic mice develop insulinitis and diabetes. *J. Exp. Med.* **1993**, *178*, 793–803.
14. Rasooly, L.; Burek, C.L.; Rose, N.R. Iodine-induced autoimmune thyroiditis in NOD-H-2h4 mice. *Clin. Immunol. Immunopathol.* **1996**, *81*, 287–291.
15. Goillot, E.; Mutin, M.; Touraine, J.L. Sialadenitis in nonobese diabetic mice: transfer into healthy neonates by splenic T lymphocytes: *Clin. Immunol. Immunopathol.* **1991**, *59*, 462–473.
16. Cho, K.H.; Han, S.H.; Lim, J.K.; Shon, Y.H.; Lee, Y.T.; Nam, K.S. Antitumor activity of Gamdutang aqua-acupuncture solution. *Kor. J. Life Sci.* **1999**, *9*, 677–683.
17. Han, S.H.; Cho, K.H.; Choi, H.K.; Lim, J.K.; Shon, Y.H.; Lee, Y.T. Nam, K.S. Chemopreventive effect of Gamdutang aqua-acupuncture solution. *Kor. J. Life Sci.* **1999**, *9*, 684–691.
18. Cho, K.H.; Kim, J.W.; Park, G.M.; Shon, Y.H.; Nam, K.S.; Lim, J.K. Effect of Medicinal plants aqua-acupuncture solution on tumor cells. *J. Kor. AM-Meridian and Pointology Soc.* **2000**, *17*, 7–11.
19. Cho, K.H.; Yoon, S.M.; Park, G.M.; Shon, Y.H.; Nam, K.S.; Lim, J.K. Immunomodulatory effects of aqua-acupuncture solution of medicinal plants. *J. Kor. AM-Meridian and Pointology Soc.* **2000**, *17*, 7–11.
20. Shon, Y.H.; Lee, H.S.; Lim, J.K.; Kim, C.H.; Jeon, B.H.; Nam, K.S. Effect of herbal medicines on cytokine-induced cytotoxicity and MHC class II antigen expression in rat thyroid cells. *Biol. Pharm. Bull.* **2004**, *27*, In press.
21. Mosmann, T.R.; Sad, S. The expanding universe of T-cell subsets: Th1, Th2 and more. *Immunol Today*, **1996**, *17*, 138–146.
22. Laemmli, UK. Cleavage of structural proteins during the assembly of the head of bacteriophage T4. *Nature.* **1970**, *227*, 680–685.
23. Kim, C.H.; Sata, H.; Taniguchi, H.; Maruyama, Y. Cloning and expression of raw-starch-digesting alpha-amylase gene from *Bacillus circulans* F-2 in *Escherichia coli*. *Biochim. Biophys. Acta.* **1990**, *1048*, 223–230.
24. Hawkes, R. Identification of concanavalin A-binding proteins after sodium dodecyl sulfate–gel electrophoresis and protein blotting. *Anal. Biochem.* **1982**, *123*, 143–146.
25. Trimble, R.B.; Maley, F. Optimizing hydrolysis of N-linked high-mannose oligosaccharides by endo-beta-N-acetylglucosaminidase H. *Anal. Biochem.* **1984**, *141*, 515–522.
26. Edge, A.S.; Faltynek, C.R.; Hof, L.; Reichert, L.E. Jr.; Weber, P. Deglycosylation of glycoproteins by trifluoromethanesulfonic acid. *Anal. Biochem.* **1981**, *118*, 131–137.
27. Champion, B.R.; Page, K.R.; Parish, N.; Rayner, D.C.; Dawe, K.; Biswas-Hughes, G.; Cooke, A.; Geysen, M.; Roitt, J.M. Identification of a thyroxine-containing

- self-epitope of thyroglobulin which triggers thyroid autoreactive T cells. *J. Exp. Med.* **1991**, 174, 363–370.
28. Engvall, E.; Perlmann, P. Enzyme-linked immunosorbent assay (ELISA): quantitative assay of immunoglobulin G. *Immunochemistry.* **1971**, 8, 871–876.
 29. Harach, H.R.; Williams, E.D. Thyroid cancer and thyroiditis in the goitrous region of Salta, Argentina, before and after iodine prophylaxis. *Clin. Endocrinol.* **1995**, 43, 701–706.
 30. Herzog, V.; Berndorfer, U.; Saber, Y. Isolation of insoluble secretory product from bovine thyroid: extracellular storage of thyroglobulin in covalently cross-linked form. *J Cell Biol.* **1992**, 118, 1071–83.
 31. Ericson, L.E.; Ring, K.M.; Ofverholm, T. Selective macropinocytosis of thyroglobulin in rat thyroid follicles. *Endocrinology.* **1983**, 113, 1746–53.
 32. Mattern, T.; Flad, H.D.; Brade, L.; Rietschel, E.T.; Ulmer, A.J. Stimulation of human T lymphocytes by LPS is MHC unrestricted, but strongly dependent on B7 interactions. *J. Immunol.* **1998**, 160, 3412–3417.
 33. Shah, P.D.; Gilbertson, S.M.; Rowley, D.A. Dendritic cells that have interacted with antigen are targets for NK cells. *J. Exp. Med.* **1985**, 162, 625–630.
 34. Karpus, W.J.; Lukacs, N.W.; McRae, B.L.; Strieter, R.M.; Kunkel, S.L.; Miller, S.D. An important role for the chemokine macrophage inflammatory protein-1 in the pathogenesis of the T cell-mediated autoimmune disease, experimental autoimmune encephalomyelitis. *J. Immunol.* **1995**, 155, 5003–5009.
 35. Springer, T.A. Traffic signals for lymphocyte recirculation and leukocyte emigration: the multistep paradigm. *Cell.* **1994**, 76, 301–308.